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Chapter 1: Landing Page & Public Features

Pass	#	Test Step	Expected Result
[]	1	Open the landing page URL in a browser	Page loads within 3 seconds; HTTP 200 response; no console errors
[]	2	Visually verify all sections are present on the page	All 10+ sections visible: Hero, How It Works, Features, Smart Matching, Trust & Safety, Pricing, Testimonials, Footer, etc.
[]	3	Click each navigation link: How It Works, Features, Smart Matching, Trust & Safety, Pricing, Testimonials	Page smoothly scrolls to the corresponding section; section heading is visible in the viewport
[]	4	Click the Login button in the navigation bar	Redirected to the login page (/login)
[]	5	Click the Get Started button	Redirected to the registration page (/register)
[]	6	Resize browser to 375px width (mobile) and verify layout	All sections display correctly without horizontal overflow; navigation collapses to hamburger menu if applicable; text is readable
[]	7	Resize browser to 768px width (tablet) and verify layout	Two-column layouts adapt appropriately; images and cards resize without overflow
[]	8	Resize browser to 1440px width (desktop) and verify layout	Full desktop layout with maximum content width; adequate side margins on large screens
[]	9	Check browser DevTools for PWA manifest and icon files	manifest.json loads successfully; icon-192.png returns HTTP 200; icon-512.png returns HTTP 200

Pass	#	Test Step	Expected Result
[]	10	Scroll to the footer and verify all links and content	Footer is visible with all links (Privacy Policy, Terms, Contact, etc.); social media links are present and functional

Chapter 2: Authentication System

2.1 Email Registration

Pass	#	Test Step	Expected Result
[]	1	Navigate to /register and select Family Member role; fill all fields (name, email, phone, password) and submit	Account created successfully; user redirected to OTP verification page; success message displayed
[]	2	Navigate to /register and select Caregiver role; fill all fields and submit	Account created successfully; user redirected to OTP verification page
[]	3	Attempt registration with an email that already exists in the database	Error message displayed: email already registered; form does not submit
[]	4	Submit registration with password shorter than minimum length (e.g., 4 characters)	Validation error: password too short; form does not submit
[]	5	Submit registration with an invalid phone number format	Validation error: invalid phone number; form does not submit
[]	6	After successful registration, check the EmailLog table or backend console for OTP email	EmailLog table contains a new row with the registered email; email body contains a 6-digit OTP
[]	7	Enter the correct OTP on the verification page and submit	OTP verified; user redirected to the appropriate dashboard based on role; emailVerified flag set to true

2.2 Email Login

Pass	#	Test Step	Expected Result
[]	1	Navigate to /login; enter valid email and password; submit	User authenticated; redirected to the role-appropriate dashboard; session cookie is set
[]	2	Enter valid email with incorrect password; submit	Error message displayed: invalid credentials; user stays on login page
[]	3	Enter an email that is not registered; submit	Error message displayed: user not found; user stays on login page
[]	4	After successful login, refresh the page (F5)	User remains logged in; session persists; dashboard loads without re-authentication

2.3 Google OAuth

Pass	#	Test Step	Expected Result
[]	1	Navigate to /login and verify the Google OAuth button is visible	Google sign-in button rendered on the login page with correct branding/icon
[]	2	Click the Google OAuth button	Browser redirects to Google consent screen; correct client ID and redirect URI are used
[]	3	Complete Google authentication (select or create account)	User redirected back to the application; user record created (if new) or updated (if existing); session cookie set
[]	4	After OAuth redirect, verify session state	User is logged in; correct user data (name, email, avatar) populated; dashboard accessible

2.4 Phone OTP Verification

Pass	#	Test Step	Expected Result
[]	1	Trigger phone OTP send from the verification endpoint; check response body in dev mode	Response includes devOtp field in development mode; HTTP 200 returned
[]	2	Enter the correct 6-digit OTP and submit	Phone verified successfully; phoneVerified flag set to true in user record
[]	3	Enter an incorrect 6-digit OTP and submit	Error message displayed: invalid OTP; phoneVerified flag remains false
[]	4	Wait 5 minutes after OTP is generated, then submit the same OTP	Error message displayed: OTP expired; user can request a new OTP
[]	5	After successful phone verification, check the user record in the database	phoneVerified field is true; updated timestamp reflects the verification time

2.5 Password Management

Pass	#	Test Step	Expected Result
[]	1	Navigate to Change Password; enter current password, new password, confirm new password; submit	Password changed successfully; user can log in with the new password
[]	2	Enter incorrect current password in the Change Password form	Error message displayed: current password is incorrect; password not changed
[]	3	Navigate to Forgot Password; enter registered email to trigger OTP	OTP email sent to the registered address; EmailLog records the email
[]	4	Enter correct OTP, then set a new password	Password reset successfully; user can log in with the new password; old password no longer works

Chapter 3: Family Member Dashboard

3.1 Patient Management

Pass	#	Test Step	Expected Result
[]	1	Navigate to Patient Management; click Add Patient; fill all fields (name, age, condition, care needs, etc.) and submit	Patient profile created; appears in patient list with correct details
[]	2	Navigate to Patient List and verify all patients display	Patient list shows all created patients with name, age, and condition summary
[]	3	Click on a patient from the list to view details	Patient detail page shows all fields: name, age, medical conditions, care requirements, assigned caregivers
[]	4	Click Edit on a patient profile; modify one or more fields; save	Patient profile updated; changes reflected on the detail page and list view

3.2 Caregiver Search & Smart Matching

Pass	#	Test Step	Expected Result
[]	1	Navigate to Caregiver Search; enter a city name and search	Results show only caregivers whose city matches the search term
[]	2	Apply filter by skill (e.g., "Elderly Care")	Results narrow to caregivers who have the specified skill in their profile
[]	3	Apply filter by verification status (verified/unverified)	Results update to show only caregivers matching the selected verification status
[]	4	Apply filter by minimum rating (e.g., 4 stars)	Results show only caregivers with average rating $\geq$ the specified value
[]	5	Navigate to Smart Matching; select care requirements (condition type, shift preference, etc.) and run matching	Results display caregivers with weighted match scores; scores reflect how well caregivers match the requirements
[]	6	Click on a caregiver from search or matching results to view their full profile	Caregiver detail page shows: name, photo, skills, experience, ratings, reviews, availability, hourly rate, verification badges

3.3 Booking Management

Pass	#	Test Step	Expected Result
[]	1	Navigate to Create Booking; select a patient, caregiver, shift type (day/night), start/end dates, and times; submit	Booking created with status "pending"; all details saved correctly in the database
[]	2	Verify the cost calculation on a new booking (hourly rate x hours per shift x number of days + 10% platform fee)	Total cost matches the formula: (hourly_rate x hours x days) x 1.10; displayed cost is correct
[]	3	Navigate to Booking List; use status filters (pending, confirmed, in-progress, completed, cancelled)	List updates to show only bookings matching the selected filter; filter toggle works correctly
[]	4	Click on a booking to view its detail page	Detail page shows: patient info, caregiver info, payment summary, linked care reports, and reviews
[]	5	Select a pending/confirmed booking and click Cancel; enter a cancellation reason; submit	Booking status changes to "cancelled"; cancellation reason is recorded; caregiver is notified
[]	6	Create an urgent care request and verify the 25% surcharge	Urgent booking created with total cost = standard cost x 1.25; urgent flag is visible on booking detail

3.4 Care Reports

Pass	#	Test Step	Expected Result
[]	1	Navigate to a confirmed/in-progress/completed booking and open the Care Reports section	List of care reports submitted for this booking is displayed, sorted by date (newest first)
[]	2	Click on a care report to view its details	Report shows: date, activities performed, patient mood, food intake, medicines given, any concerns flagged

3.5 Reviews

Pass	#	Test Step	Expected Result
[]	1	Navigate to a completed booking and open the Review section; submit a review with ratings for all 4 dimensions (overall, communication, punctuality, care quality) and optional comment	Review submitted successfully; all 4 ratings and comment are saved
[]	2	Navigate to the reviewed caregiver profile	The new review appears on the caregiver profile with correct ratings and comment; average rating updates

3.6 Complaints

Pass	#	Test Step	Expected Result
[]	1	Navigate to Complaints section; click File Complaint; fill subject, description, and priority; submit	Complaint created with status "open"; confirmation message displayed
[]	2	View the complaints list and check status updates	Each complaint shows current status (open, in-progress, resolved); status history/timeline is visible

3.7 Notifications

Pass	#	Test Step	Expected Result
[]	1	Verify the notification bell icon in the header shows an unread count badge	Bell icon displays a numeric badge when there are unread notifications; badge count matches actual unread count
[]	2	Click the bell icon to open the notification dropdown/list	Notification list appears showing all notifications with timestamps, read/unread state, and brief descriptions
[]	3	View a notification; verify it is automatically marked as read	Opened notification status changes to read; unread count badge decreases by 1; bell badge updates

Chapter 4: Caregiver Dashboard

4.1 Profile Setup

Pass	#	Test Step	Expected Result
[]	1	Navigate to Profile Setup; fill all fields (city, gender, address, experience years, skills, bio, hourly rate); submit	Profile saved successfully; all fields reflect the entered values on the profile page
[]	2	Navigate to Availability; set day shift and night shift availability for each day of the week; save	Weekly availability schedule saved; correct shift types displayed for each day; available time slots match settings

4.2 Document Verification

Pass	#	Test Step	Expected Result
[]	1	Navigate to Document Upload; select a document type (e.g., Aadhaar, PAN); upload a valid file; submit	Document uploaded successfully; appears in document list with type, file name, and status "pending"
[]	2	Upload an Aadhaar card image and verify AI OCR extraction	System extracts name, date of birth, Aadhaar number from the image; extracted data is displayed for review before submission
[]	3	Check the verification status of an uploaded document	Document shows correct status: pending, approved, or rejected; rejection reason visible if rejected

4.3 Booking Management

Pass	#	Test Step	Expected Result
[]	1	Navigate to My Bookings; view the list of assigned bookings	All bookings assigned to this caregiver are listed with status, patient name, dates, and shift type
[]	2	Select a pending booking and click Confirm	Booking status changes to "confirmed"; family member receives a notification
[]	3	Select a confirmed booking and click Mark In-Progress	Booking status changes to "in-progress"; care report submission becomes available
[]	4	Select an in-progress booking and click Mark Completed	Booking status changes to "completed"; family member is notified; review submission becomes available

4.4 Care Reports

Pass	#	Test Step	Expected Result
[]	1	For an in-progress booking, navigate to Submit Care Report; fill all fields (activities, mood, food, medicines, concerns); submit	Care report created successfully; report appears in the booking care reports section
[]	2	After submitting a care report, check the family member notification	Family member receives a real-time notification (Socket.io) about the new care report
[]	3	After submitting a care report, check the EmailLog table for notification email	EmailLog contains a new entry for the family member; email content includes care report summary with highlighted concerns

4.5 Earnings & Withdrawals

Pass	#	Test Step	Expected Result
[]	1	Navigate to Earnings section and view earnings history	Earnings page shows a list of all completed booking payments with date, amount, and booking reference
[]	2	Navigate to Withdrawals; enter amount and select withdrawal method (UPI or bank transfer); submit	Withdrawal request created with status "pending"; available balance decreases by the requested amount
[]	3	Verify the balance calculation: available = 90% of total earnings minus pending withdrawal amounts	Displayed available balance matches: $(total_earned \times 0.90) - sum(pending_withdrawals)$; calculation is accurate

4.6 Reviews

Pass	#	Test Step	Expected Result
[]	1	Navigate to My Reviews; view the list of received reviews	All reviews from family members are displayed with ratings (overall, communication, punctuality, care quality), comments, and dates

Chapter 5: Admin Dashboard

5.1 Overview Analytics

Pass	#	Test Step	Expected Result
[]	1	Navigate to Admin Dashboard; verify all stat cards display correctly	Cards show: total users, total bookings, total revenue, average rating, open complaints; all values are non-negative numbers
[]	2	Verify the Bookings by Status chart renders	Chart displays with correct segments for each booking status (pending, confirmed, in-progress, completed, cancelled); legend is readable
[]	3	Verify the Complaints by Status chart renders	Chart displays with correct segments for complaint statuses; data matches database counts
[]	4	Verify the Caregivers by City chart renders	Chart displays caregiver distribution across cities; city names are readable; counts are accurate

5.2 User Management

Pass	#	Test Step	Expected Result
[]	1	Navigate to User Management; search by name	Results show users whose name contains the search term; results are accurate
[]	2	Search users by email	Results show the user matching the email exactly; partial email search works if implemented
[]	3	Search users by phone number	Results show the user matching the phone number
[]	4	Apply role filter (Family, Caregiver, Admin)	User list updates to show only users with the selected role
[]	5	Click on a user to view detailed profile	User detail page shows: name, email, phone, role, registration date, verification status, associated patients/bookings
[]	6	Delete a test user (non-critical account) and verify cascade delete	User removed from database; associated records (patients, bookings for test user) are also removed or handled gracefully

5.3 Verification Management

Pass	#	Test Step	Expected Result
[]	1	Navigate to Verification Management; view the list of verification documents with filters	List shows all uploaded documents with caregiver name, document type, upload date, and status
[]	2	Apply filters (document type, status, caregiver name)	List updates correctly based on applied filters; combined filters work together
[]	3	Select a pending document and click Approve	Document status changes to "approved"; caregiver is notified
[]	4	Select a pending document and click Reject; enter a rejection reason; submit	Document status changes to "rejected"; rejection reason is recorded and visible; caregiver is notified
[]	5	Approve all required documents for a caregiver and verify auto-promotion	When all document types are approved, caregiver verification status automatically changes to "verified"

5.4 Booking Management

Pass	#	Test Step	Expected Result
[]	1	Navigate to Admin Booking Management; view all platform bookings	List displays all bookings across the platform with patient name, caregiver name, dates, status, and amount
[]	2	Search bookings by patient name or caregiver name	Results show bookings matching the search term

5.5 Withdrawal Management

Pass	#	Test Step	Expected Result
[]	1	Navigate to Withdrawal Management; view all withdrawal requests	List shows all withdrawal requests with caregiver name, amount, method (UPI/bank), date, and status
[]	2	Select a pending withdrawal and process it through the status workflow (pending -> processing -> completed)	Status updates correctly at each step; caregiver notified upon completion
[]	3	Select a pending withdrawal and click Reject; enter a reason	Status changes to "rejected"; reason is recorded; caregiver balance is restored

5.6 Review & Complaint Management

Pass	#	Test Step	Expected Result
[]	1	Navigate to Review Management; view all reviews across the platform	List shows all reviews with reviewer name, caregiver name, ratings, date, and comment
[]	2	Navigate to Complaint Management; view all complaints	List shows all complaints with complainant, subject, priority, status, and date
[]	3	Select a complaint; assign it to an admin or caregiver; update status to in-progress	Complaint assigned correctly; status changes; assigned user receives notification
[]	4	Resolve a complaint and add resolution notes	Complaint status changes to "resolved"; resolution notes are saved and visible in the complaint timeline

Chapter 6: Real-Time Features (Socket.io)

Pass	#	Test Step	Expected Result
[]	1	Verify the Socket.io service is running on port 3005	Service responds to connection requests on port 3005; no errors in server logs
[]	2	Trigger a payment status update as an admin; verify the family member receives a real-time notification	Family member browser receives a Socket.io event with payment update details; UI updates without page refresh
[]	3	Trigger a booking status change (e.g., confirm); verify the caregiver receives a real-time notification	Caregiver browser receives a Socket.io event with booking update details; UI updates without page refresh
[]	4	Submit a care report as a caregiver; verify the family member receives a real-time notification	Family member browser receives a Socket.io event for the new care report; notification appears instantly
[]	5	Disconnect the browser from the network and reconnect; verify Socket.io reconnects automatically	After reconnecting, the Socket.io connection is re-established; no missed events during disconnect; connection state recovers

Chapter 7: Email Service

Pass	#	Test Step	Expected Result
[]	1	Trigger OTP email by initiating registration or password reset	Email is sent via Gmail SMTP; email arrives in the recipient inbox (or spam); contains the correct 6-digit OTP
[]	2	Create a new booking and verify the booking confirmation email	Email sent to family member with booking details: patient name, caregiver name, dates, shift, total cost
[]	3	Submit a care report with concerns flagged; verify the care report notification email	Email sent to family member with care report summary; flagged concerns are highlighted or prominently displayed in the email body

Pass	#	Test Step	Expected Result
[]	4	After sending any email, check the EmailLog table in the database	EmailLog table contains a new row with: recipient email, subject, email type/status, timestamp; all fields are populated

Chapter 8: SMS Service

Pass	#	Test Step	Expected Result
[]	1	Trigger phone OTP send via the API endpoint	In production: SMS sent via Fast2SMS to the phone number; in dev mode: devOtp returned in the API response body
[]	2	In development mode, verify the devOtp is returned in the response	API response JSON includes a devOtp field with the 6-digit OTP; no actual SMS is sent in dev mode
[]	3	After OTP send, verify the OTP is stored in the database	Database contains the OTP record with: phone number, OTP code, expiration timestamp, used flag (initially false)

Chapter 9: API Health Check

Pass	#	Test Step	Expected Result
[]	1	Send GET/POST requests to all auth endpoints (/auth/register, /auth/login, /auth/google, /auth/otp-send, /auth/otp-verify)	All endpoints return appropriate HTTP status codes (200, 201, 400, 401); response bodies match API schema
[]	2	Send GET/POST/PUT requests to all booking endpoints (/bookings, /bookings/:id, /bookings/cancel, /bookings/urgent)	All endpoints respond correctly; CRUD operations work as expected; status transitions are enforced

Pass	#	Test Step	Expected Result
[]	3	Send GET/PUT requests to all caregiver endpoints (/caregivers, /caregivers/:id, /caregivers/profile, /caregivers/availability)	All endpoints respond with correct data; profile updates persist; availability schedules save correctly
[]	4	Send GET requests to all admin endpoints (/admin/users, /admin/verifications, /admin/bookings, /admin/withdrawals, /admin/reviews, /admin/complaints)	All endpoints respond correctly for admin users; non-admin users receive 401 or 403 Forbidden
[]	5	Send requests with invalid inputs (missing fields, wrong data types, invalid IDs)	Endpoints return HTTP 400 with descriptive error messages; no server crashes or 500 errors
[]	6	Send requests without authentication to protected endpoints	Endpoints return HTTP 401 Unauthorized; no data is leaked in the response body